Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
Fall 2025

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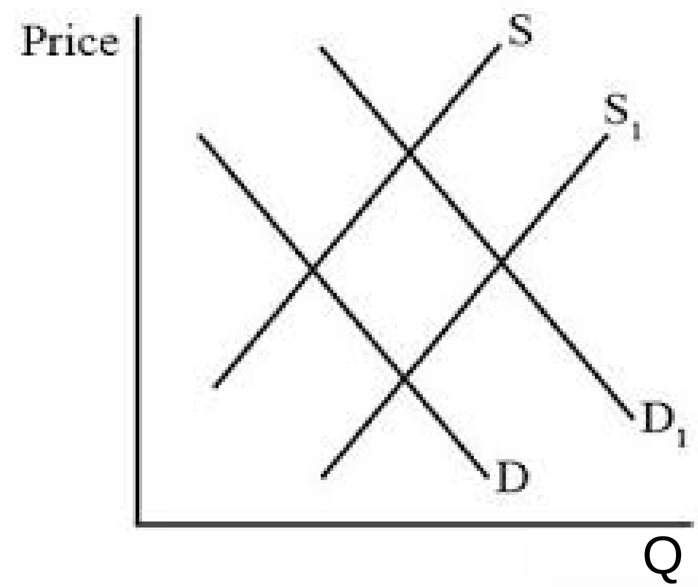
The Determinants of Interest Rates

(Chapter 4)



Outline of Lecture

Topics of today

1. Asset demand
    - determinants

2. Bond market:
    - demand curve: derivation
    - supply curve: derivation
    - equilibrium
    - shifts in demand
    - shifts in supply

Determinants of (Financial) Asset Demand

1. **Wealth**: total resources owned by the individual (house, securities, car…).

2. **Expected return** (over the next period) on one asset *relative* to alternative assets

3. **Risk** (uncertainty on asset return) on one asset *relative* to alternative assets

4. **Liquidity** (the ease and speed with which an asset can be turned into cash) *relative* to alternative assets

Wealth

- ✓ Wealth has a positive effect on the demand of financial assets
  - Note: *wealth≠income*, it's a *stock variable*, not a *flow*

- ✓ Sounds pretty obvious, since financial assets are *normal goods* (*i.e.* not *inferior goods*)

Holding everything else constant ….

an **increase** in **wealth**

increases the **demand** of an asset

Expected Return

- ✓ The **expected return** is the average return across all *states of nature*

$$E(R) = \sum_{i=1}^{N} p_i \times R_i$$

where:

R_i is the return in state i

p_i is the probability of state i occurring (technically, this is the formula for the expected value of a *discrete* random variable)

For example, consider a coin toss: you win \$1 if heads and \$0 if tails.

Q: What are p_i, R_i and $E(R)$ in this case?

Expected Return

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For example, consider a coin toss: you win $1 if heads and $0 if tails.

Q: What are p_i, R_i and $E(R)$ in this case?

$$p_H = p_T = 0.5 \; ; \; R_H = \$1, R_T = \$0$$
$$E(R) = 0.5 \times 1 + 0.5 \times 0 = \$\mathbf{0.5}$$

Expected Return

- ✓ Nice thing about expected returns is that they are **additive**
- ✓ The expected return of a *portfolio of 2 fin. assets* (*A* and *B)* is the sum of the expected returns of the two *fin. assets*:

$$E\left(R_{portfolio}\right) = E(R_A + R_B) = E(R_A) + E(R_B)$$

Expected Return

Holding <u>everything else constant</u>,
an **increase** in the **expected return**
increases the **demand** of an asset

- ✓ Again: quite intuitive!
- ✓ Note: <u>*ceteris paribus* condition</u> (= *everything else constant*) here <u>is key</u>
  - you may very well prefer an asset with lower expected return, but other factors different (e.g.: lower **risk!**)
- ✓ Often, high expected returns assets also have high risk (risk-return trade-off)

Risk

- ✓ We measure **risk** as the <u>standard deviation</u> of an asset return (across all states of the world)

$$\sigma(R) = \sqrt{\sum_{i=1}^{N} p_i \times (R_i - E(R))^2}$$

- ✓ Consider stock A and its forecasted returns for the upcoming year:

$$R_i = \begin{cases} 3\% & p = 1/4 \\ 7\% & p = 1/2 \\ 15\% & p = 1/4 \end{cases}$$

Q: what is the risk of stock A?

A: 1st: need to calculate the expected return:

$$E(R) = (0.25 \cdot 0.03) + (0.5 \cdot 0.07) + (0.25 \cdot 0.14) = \$0.08 \text{ (read } \mathbf{8}\%\text{)}$$

...

Risk: Answer

… 2nd: take square of deviations from the mean weighting by their probability:

$$\sum_{i=1}^{N} p_i \times (R_i - E(R))^2$$
$$= 0.25 \cdot (0.03 - 0.08)^2 + 0.5 \cdot (0.07)^2 + 0.25 \cdot (0.14 - -0.08)^2 = 0.19\%$$

3rd: take the square root: $\sigma = \sqrt{0.19}\% = 4.36\%$

Now consider stock B:

$$R_i = \{\mathbf{8}\% \;\; p = 1$$

Q: What is the risk of stock B?

Risk: Answer

… 2nd: take square of deviations from the mean weighting by their probability:

$$\sum_{i=1}^{N} p_i \times (R_i - E(R))^2$$
$$= 0.25 \cdot (0.03 - 0.08)^2 + 0.5 \cdot (0.07)^2 + 0.25 \cdot (0.14 - -0.08)^2 = 0.19\%$$

3rd: take the square root: $\sigma = \sqrt{0.19}\% = 4.36\%$

Now consider stock B:

$$R_i = \{8\% \;\; p = 1$$

Q: What is the risk of stock B?

A: 0 because the probability is 1 (no uncertainty!)

Risk

So, both stocks A and B have the same expected return: 8%.

✓ Should I be indifferent between the two?

- a *risk*-***averse*** individual prefers the asset with the **lower** risk (standard deviation), for the <u>same expected return</u>
- he/she chooses stock **B** (the sure thing) to stock A (the riskier asset), because risk is costly to him/her

✓ We assume that people are *risk-averse*, especially in their financial decisions

- generally, investors on financial markets are risk averse

Holding everything else constant,
an **increase** in **risk**
decreases the **demand** of an asset

Liquidity

✓ *Definition*: Ease and speed with which an asset is turn into cash

- Need many traders (buyers and sellers) and active trades
- Intuitively: the *higher the number of traders* in the market, the **easier** it will be to **sell** the asset in any period

EX: a house is generally an illiquid asset compared to other securities (stocks and bonds)

Again intuitively:

Holding everything else constant,
an **increase** in **liquidity**
increases the **demand** of an asset

Summary

Variable	Change in Variable	Change in Quantity Demanded
Wealth	↑	↑
Expected return relative to other assets	↑	↑
Risk relative to other assets	↑	↓
Liquidity relative to other assets	↑	↑

Note: Only increases in the variables are shown. The effect of decreases in the variables on the change in quantity demanded would be the opposite of those indicated in the far-right column.

Supply & Demand in the Bond Market

Definition of bond

- ✓ A bond is a debt security that promises the holder the payment of a specified amount (the face value) in the future upon maturity, as well as it may also promise the payment of periodic coupon interest
  - if the terms of repayment are not met by the bond issuer (e.g.: default), the bond holder has a claim on the assets of the bond issuer

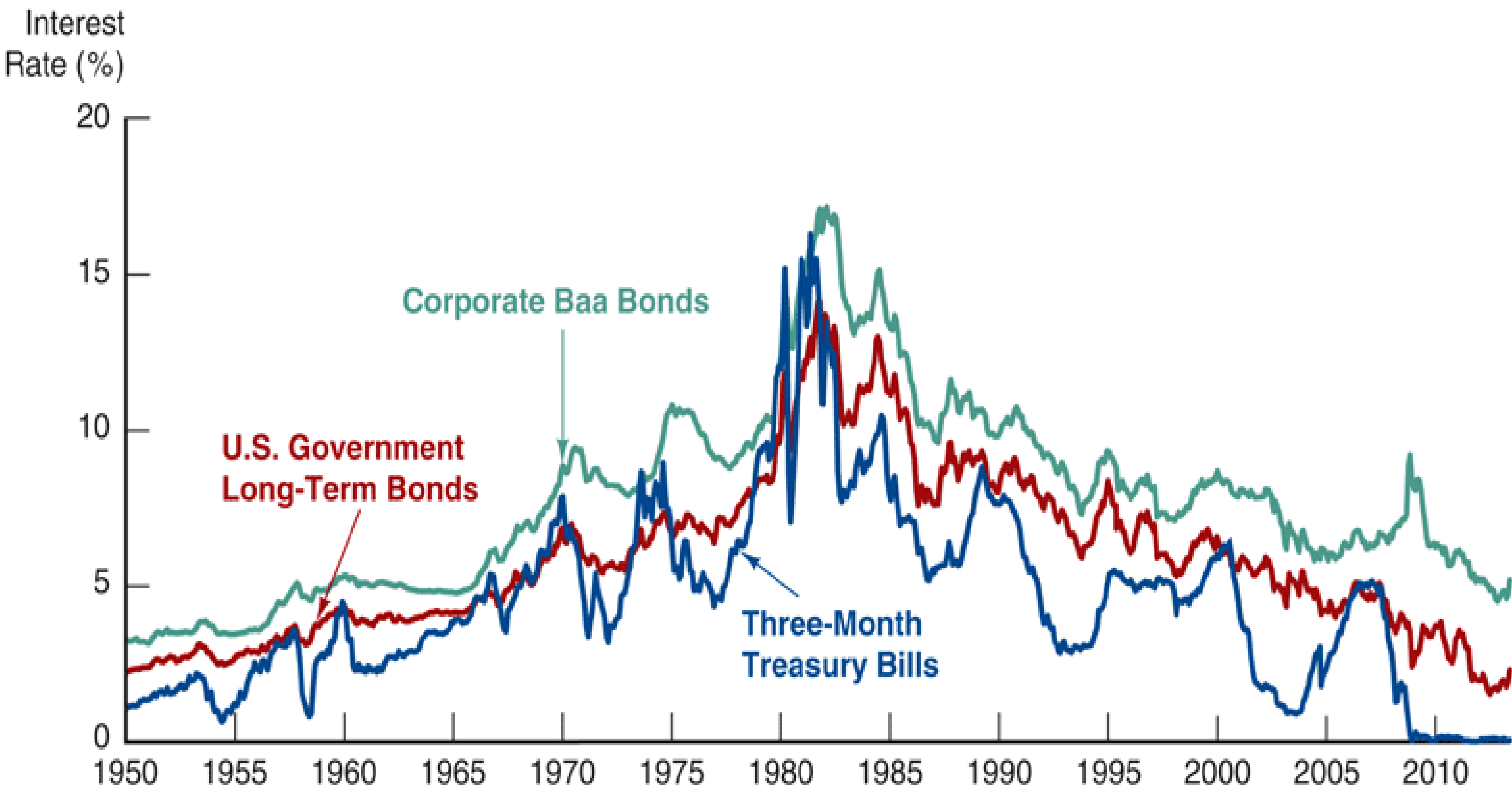
- ✓ Determination of interest rates happens through **demand and supply** of bonds

- ✓ Various types of bonds …
  - corporate bonds
  - government bonds
  - etc …

  … all have different interest rates

- ✓ However, because rates tend to move together, we will proceed as if there is one interest rate for the entire economy (like in macro)

Interest rates move together



Source: Federal Reserve Bank of St. Louis, FRED database: http://research.stlouisfed.org/fred2/.

Bonds: the Demand Curve

Let's start with the **demand** curve

- ✓ Let's consider a one-year ZCB with a face value of $1,000.
- ✓ In this case, the return on this ZCB is entirely determined by its price
- ✓ <u>The one-year return is, then, the ZCB's yield to maturity</u> (intuitively: YTM is yield if you hold the bond until maturity)

Derivation of Demand Curve

Formula for interest rate for one-year discount bond:

$$i^e = R^e = \frac{FV - P}{P}$$

✓ Point A: if the bond was selling for \$950 and face value = \$1,000:

$$P = \$950$$

$$i = \boxed{}$$

Assume $B^d = 100$ at point A (*i.e.* for $P = \$950$)

Derivation of Demand Curve

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$$i^e = R^e = \frac{FV - P}{P}$$

- ✓ Point A: if the bond was selling for \$950 and face value = \$1,000:

$$P = \$950$$

$$i = \frac{(\$1000 - \$950)}{\$950} = .053 = 5.3\%$$

Assume $B^d = 100$ at point A (*i.e.* for $P = \$950$)

Derivation of Demand Curve (cont.)

Point B: if the bond was selling for $900:

$$P = \$900$$

$$i = \frac{(\$1000 - \$900)}{\$900} = .111 = 11.1\%$$

$$B^d = 200$$

Q: Is the demand at point B likely to be higher or lower than at point A?

Derivation of Demand Curve (cont.)

Point B: if the bond was selling for $\$900$:

$$P = \$900$$

$$i = \frac{(\$1000 - \$900)}{\$900} = .111 = 11.1\%$$

$$B^d = 200$$

Q: Is the demand at point B likely to be higher or lower than at point A?

A: higher, because the expected return has gone up (lower P means higher R) and <u>nothing else has changed</u>. Suppose B^d at point B is $\$200$ (making up these numbers)

Derivation of Demand Curve

To continue …

Point C: $P = \$850 \quad i = 17.6\% \quad B^d = 300$

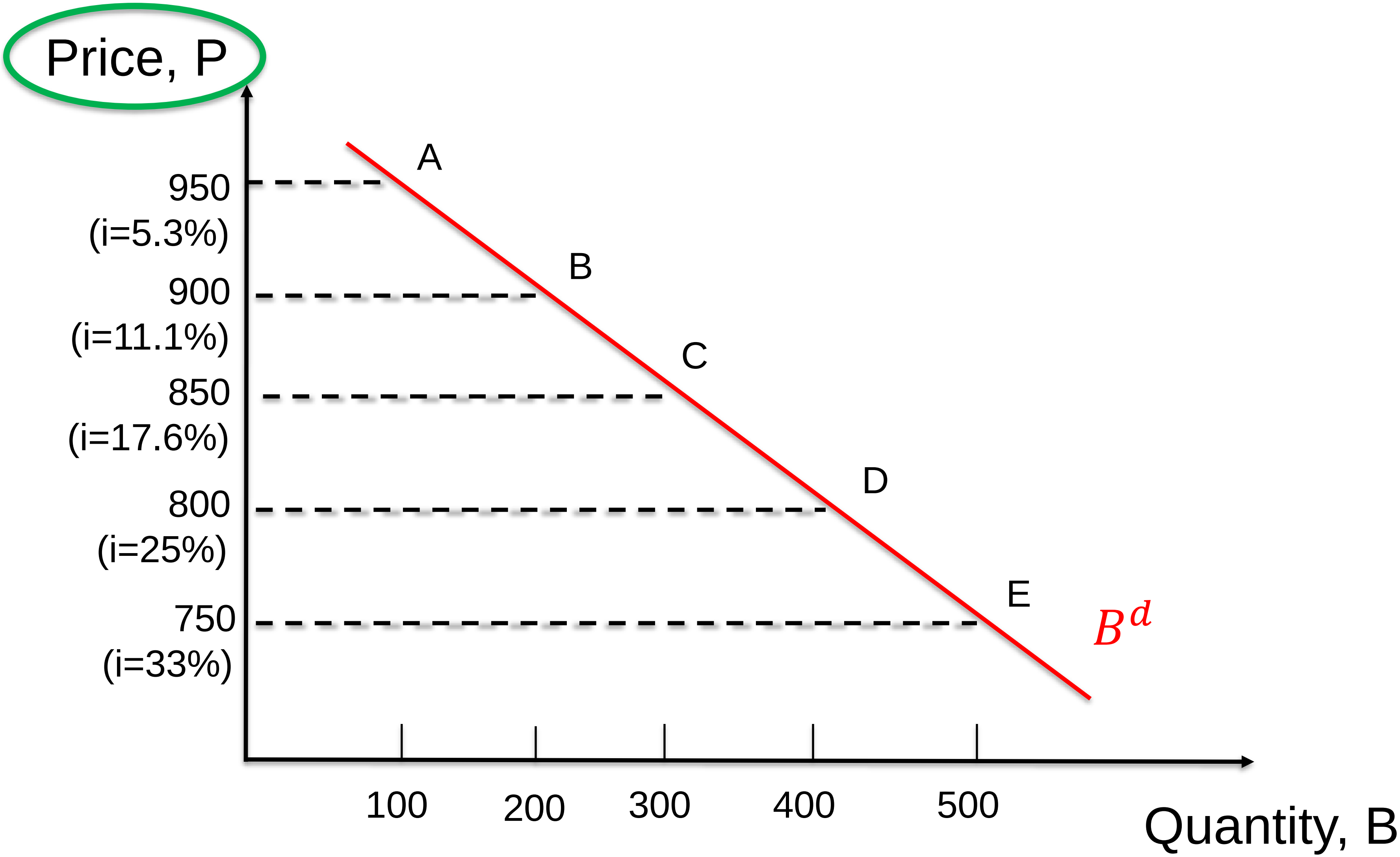
Point D: $P = \$800 \quad i = 25.0\% \quad B^d = 400$

Point E: $P = \$750 \quad i = 33.0\% \quad B^d = 500$

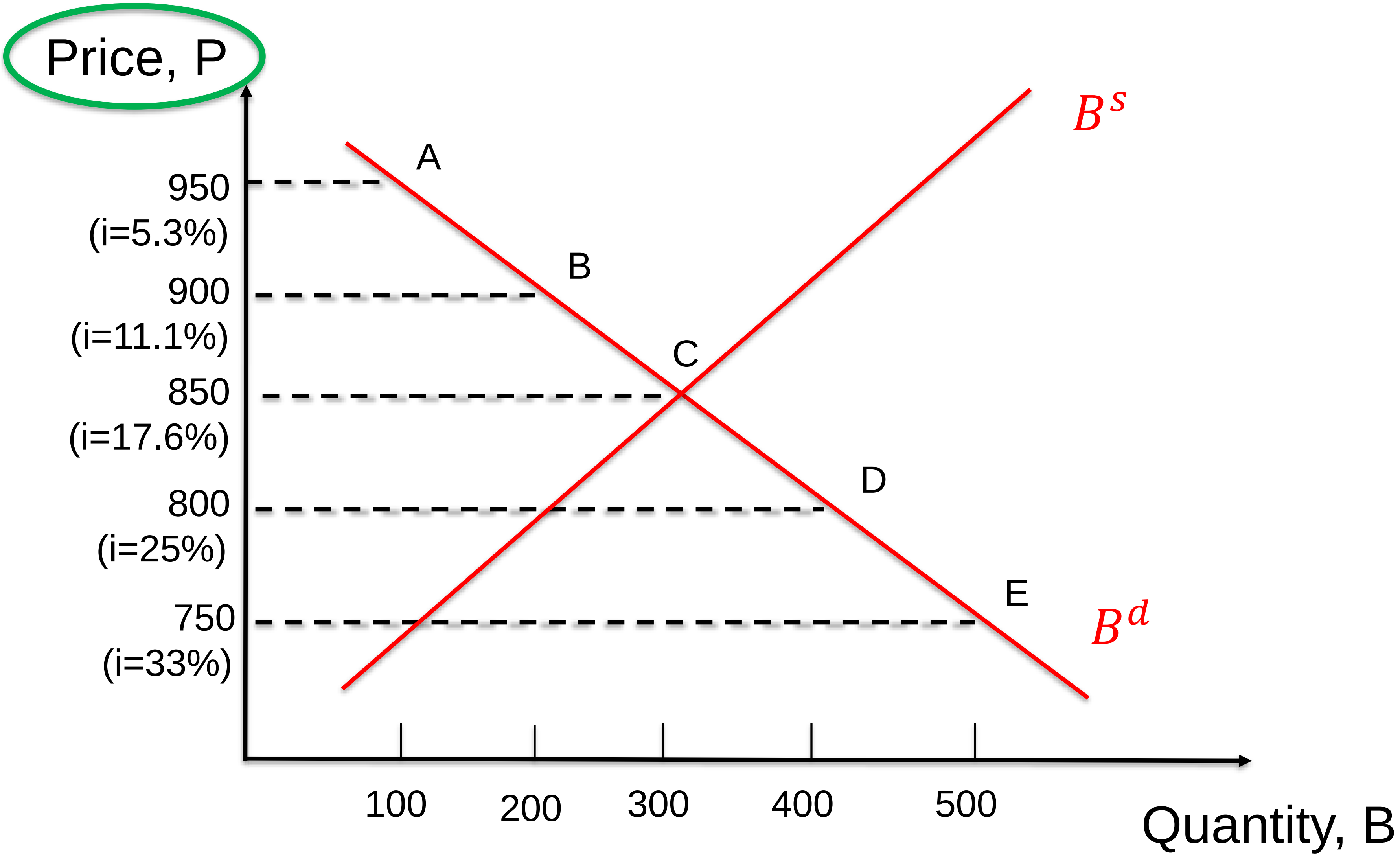
✓ As $\boldsymbol{P} \downarrow \Rightarrow R^e \uparrow \Rightarrow \boldsymbol{B^d} \uparrow \ldots$

Demand curve has usual **downward** slope with respect to **price**

Demand Curve



Demand & Supply Curve



Derivation of Supply Curve

The derivation follows the same idea as the demand curve.

Q: why is the supply curve of bonds **upward** sloping?

Derivation of Supply Curve

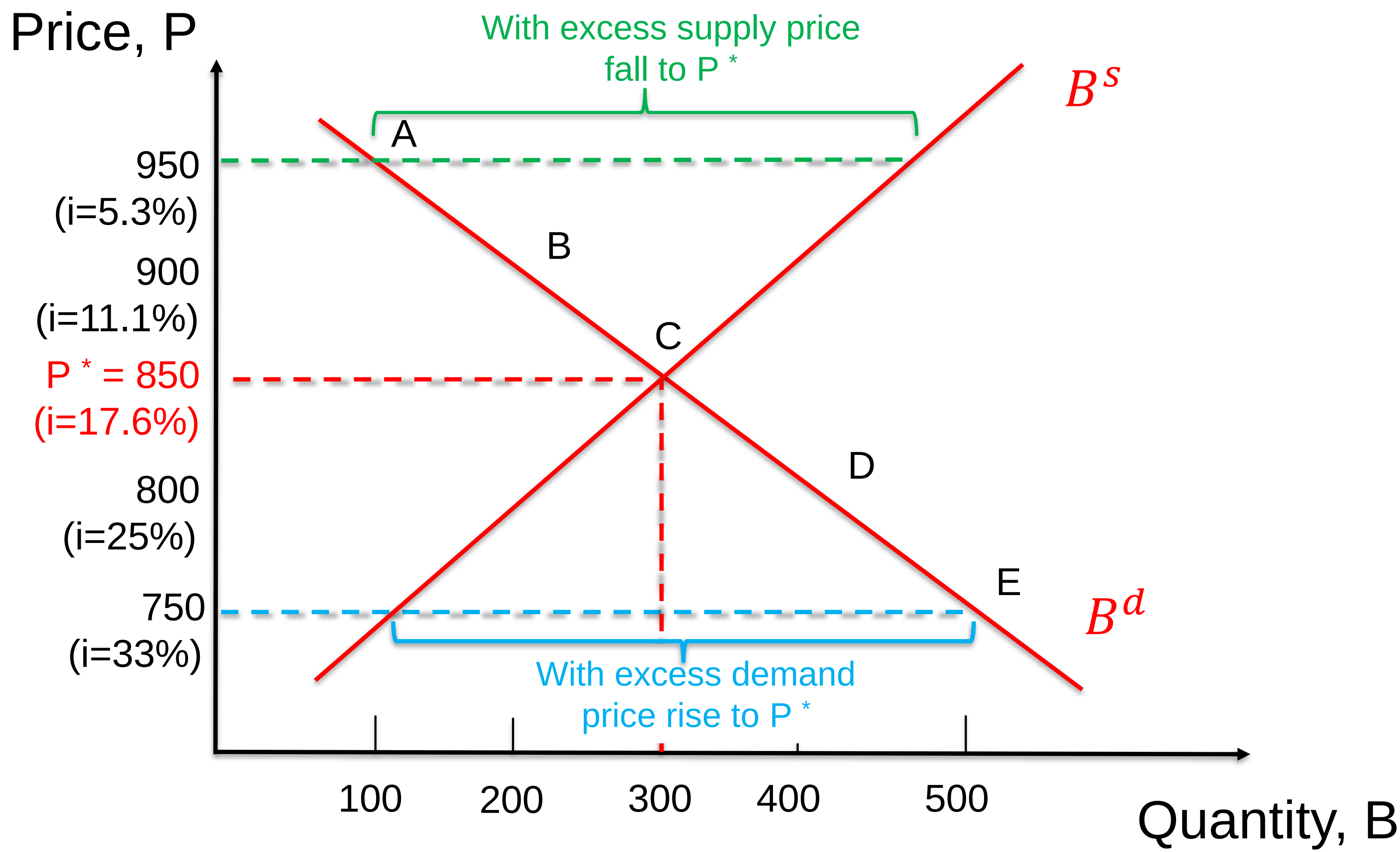
The derivation follows the same idea as the demand curve.

Q: why is the supply curve of bonds **upward** sloping?

A: If the borrowing cost for the issuer (=interest rate, i) is lower ⇒ the price is higher

- then it is more convenient for the supplier to issue more bonds
- recall supplier of bonds is a borrower (buyer is the lender)!
- more bonds will be offered (supply) if the expected return (borrowing **cost** for the **issuer**) is *lower*

Demand & Supply Curve: Equilibrium



Market Equilibrium

- ✓ **Market equilibrium** occurs when the amount that people are willing to buy (*demand*) equals the amount that people are willing to sell (*supply*) at a given price

- ✓ **Excess supply** (price is “too high”) and **excess demand** (price is “too low”) are only temporary conditions in a *competitive market …*

- ✓ <u>Market forces</u> will tend to
  - <u>equate quantities demanded</u> and <u>supplied</u>
  - <u>at the equilibrium price/interest rate</u> (*market clearing*[1])

(1): clearing the market from excess supply or demand

Market Equilibrium

The equilibrium occurs when supply-demand are equalized:

- ✓ $B^d = B^s, at\ P^* = 850, i^* = 17.6\%$
- ✓ when $P = \$950,\ i = 5.3\%,\ B^s > Bd$
  (excess **supply**): $P\downarrow to\ P^*(or\ i \uparrow to\ i^*)$
- ✓ when $P = \$750,\ i = 33.0,\ Bd > B^s$
  (excess **demand**): $P\uparrow to\ P^*\ (or\ i\downarrow to\ i^*)$

Changes in Equilibrium Interest Rates

- ✓ We now turn our attention to **shifts** in demand and in supply
- ✓ Then, we are going to put them together to derive equilibrium interest rates

Remember:

a. movements <u>*along*</u> the curves will be due to <u>price changes alone</u>

b. If **nothing else, that is not on the axes** (Q, P), changes, it's a **shift**<u> of either or both curves</u>

Shift in Demand

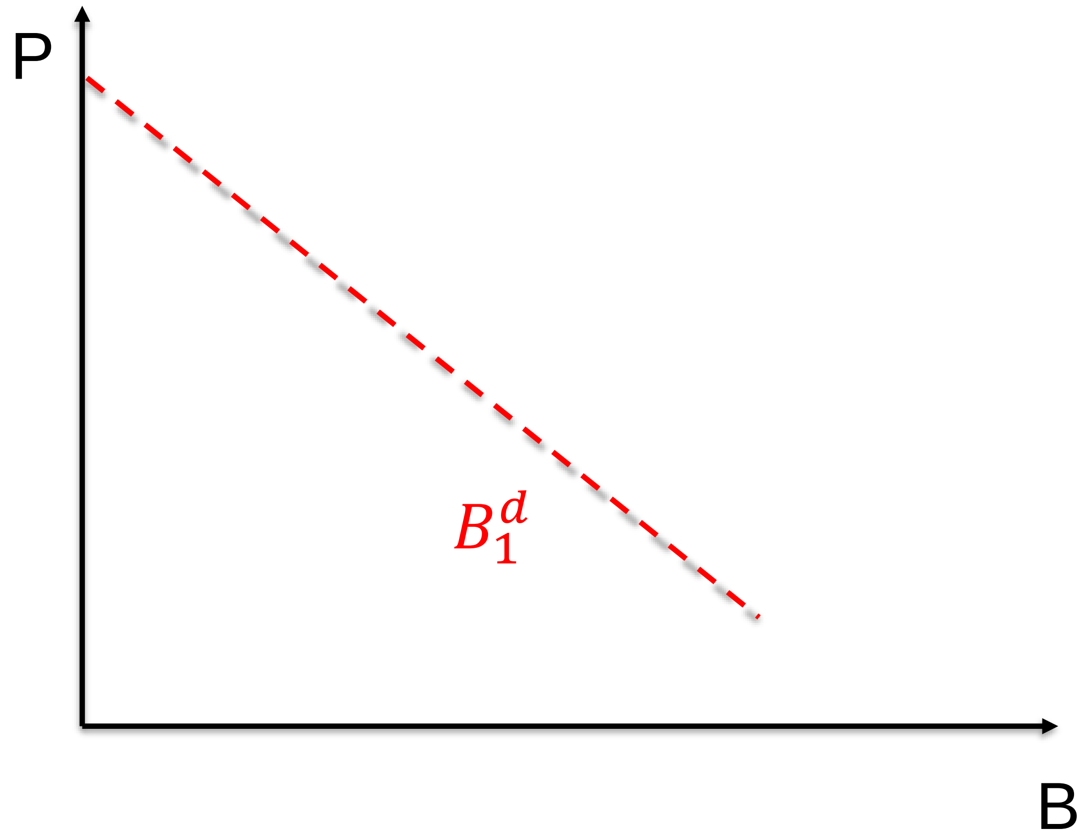
Factors that shift demand:

1. Wealth
2. Expected future interest rate: $E(i_{t+1})$
3. Riskiness: σ
4. Expected future inflation: π^e
5. Liquidity

Note: Only increases in the variables are shown. The effect |of decreases in the variables on the change in demand would be the opposite of those indicated in the remaining columns.

Shift in Demand (1)

If wealth[1] $\uparrow \Rightarrow$ demand ?
Think of economic boom (crisis) that in/de-creases wealth (abslolute)



(1): wealth may increase for example due to macro shocks, such as a *prolonged periods of expansion* (e.g.: boom); the reverse with *recession*

Shift in Demand (1)

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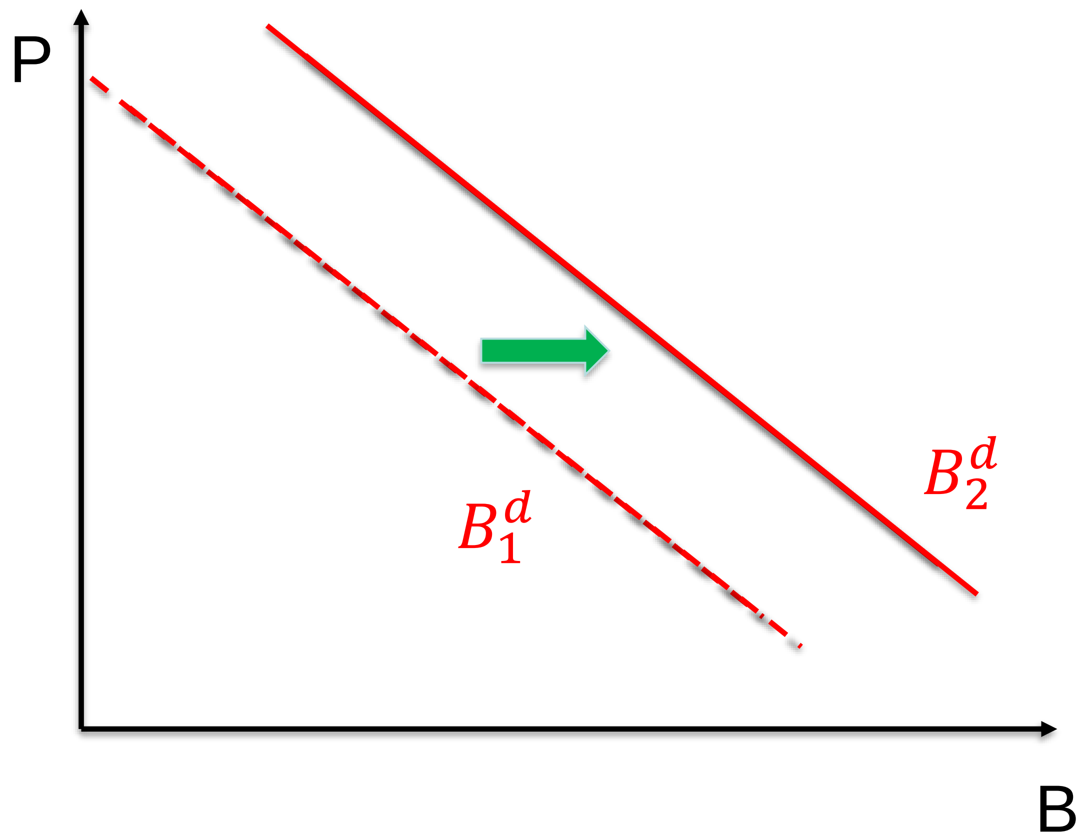
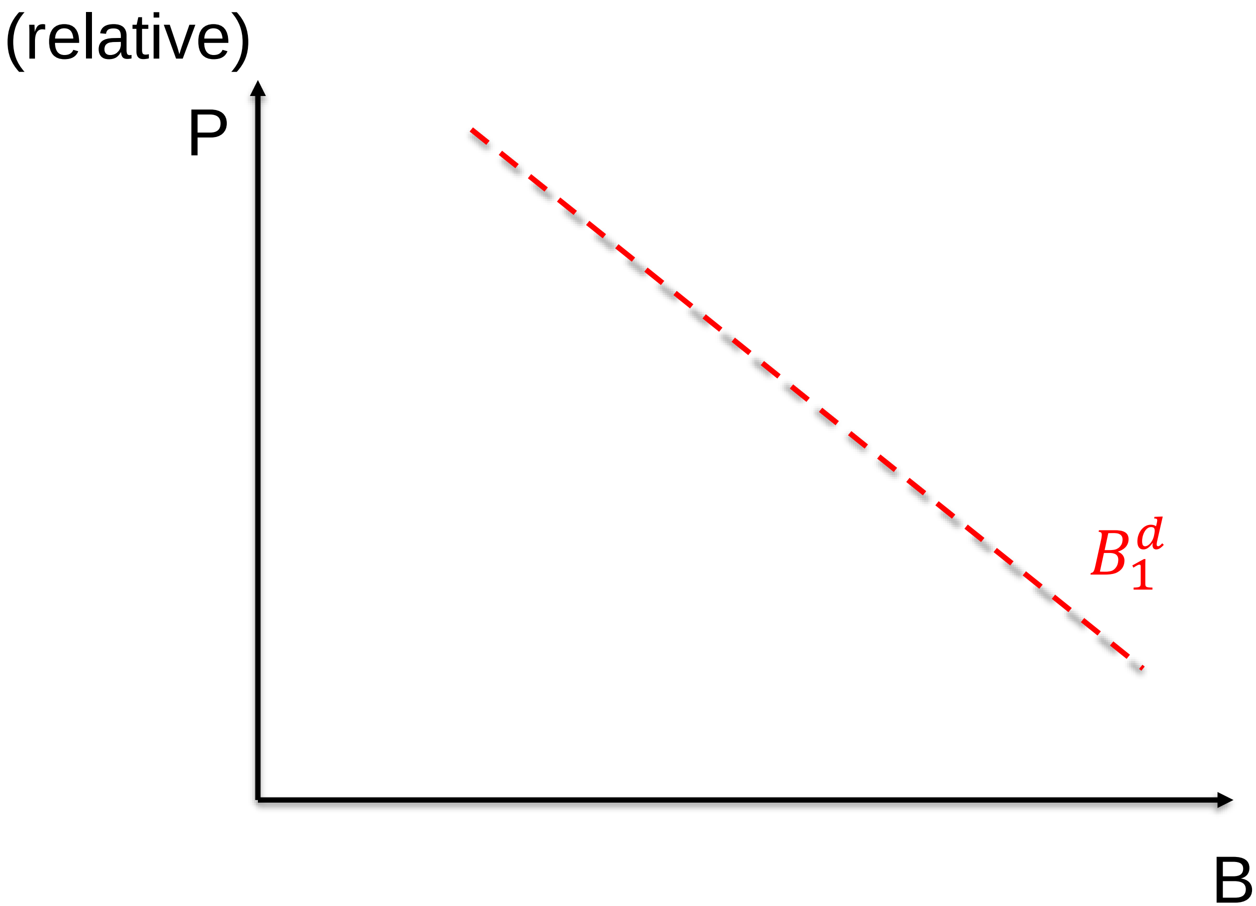


(1): wealth may increase for example due to macro shocks, such as a *prolonged periods of expansion* (e.g.: boom); the reverse with *recession*

Shift in Demand (2)

If expected interest rate ↑ in $t+1$ [$E(i_{t+1})$] ⇒ demand ?

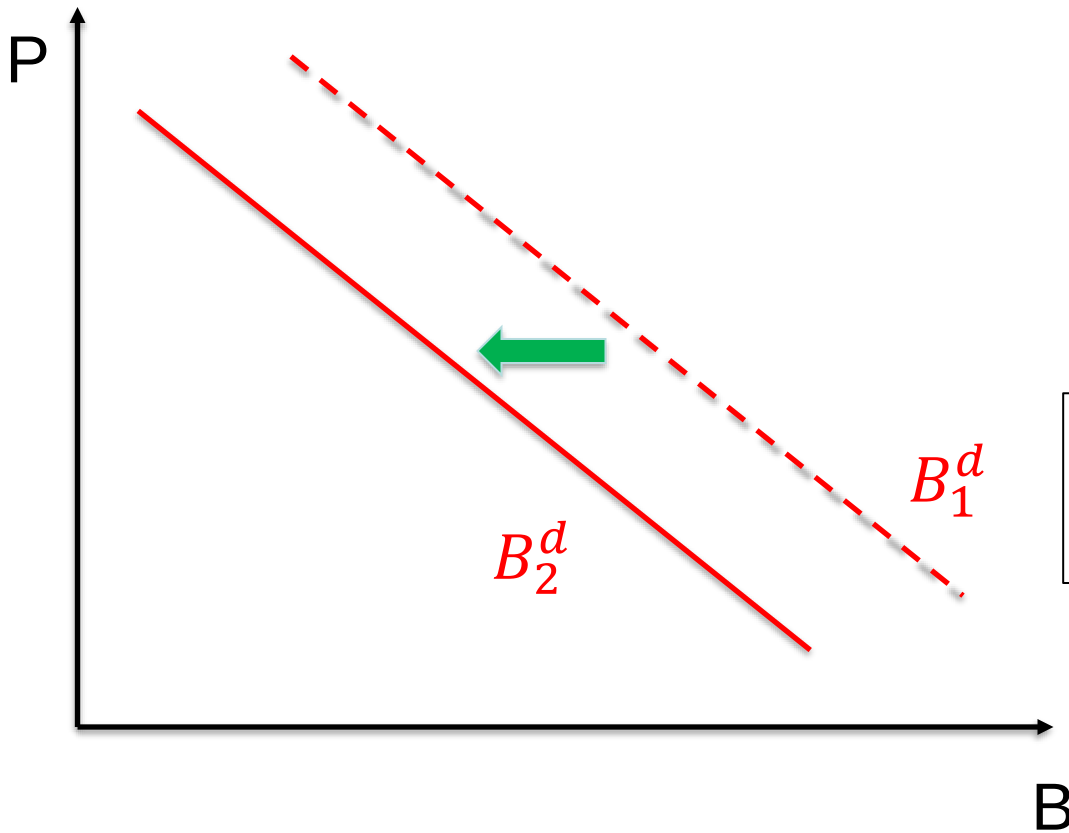


Hint: if $E(i_{t+1})$ rises what happens to **price** of **long-term** bonds $E(P_{t+1})$? Hence what happens to the **rate of return** $E(R_{t+1})$?

Q: What happens to 1-year discount bond with maturity of 1 year if $E(i)$ ↑?

Shift in Demand (2)

If expected interest rate ↑ in $t+1$ [$E(i_{t+1})$] ⇒ demand ↓ (relative)



Hint: if $E(i_{t+1})$ rises, **price** of **long**-term bonds $E(P_{t+1})$ ↓ Hence, the expected **rate of return** $E(R_{t+1})$ ↓: demand decreases

$$E(R_{t+1}) = \frac{C}{P_t} + \frac{P_{t+1} - P_t}{P_t}$$

(demand for long term bonds)

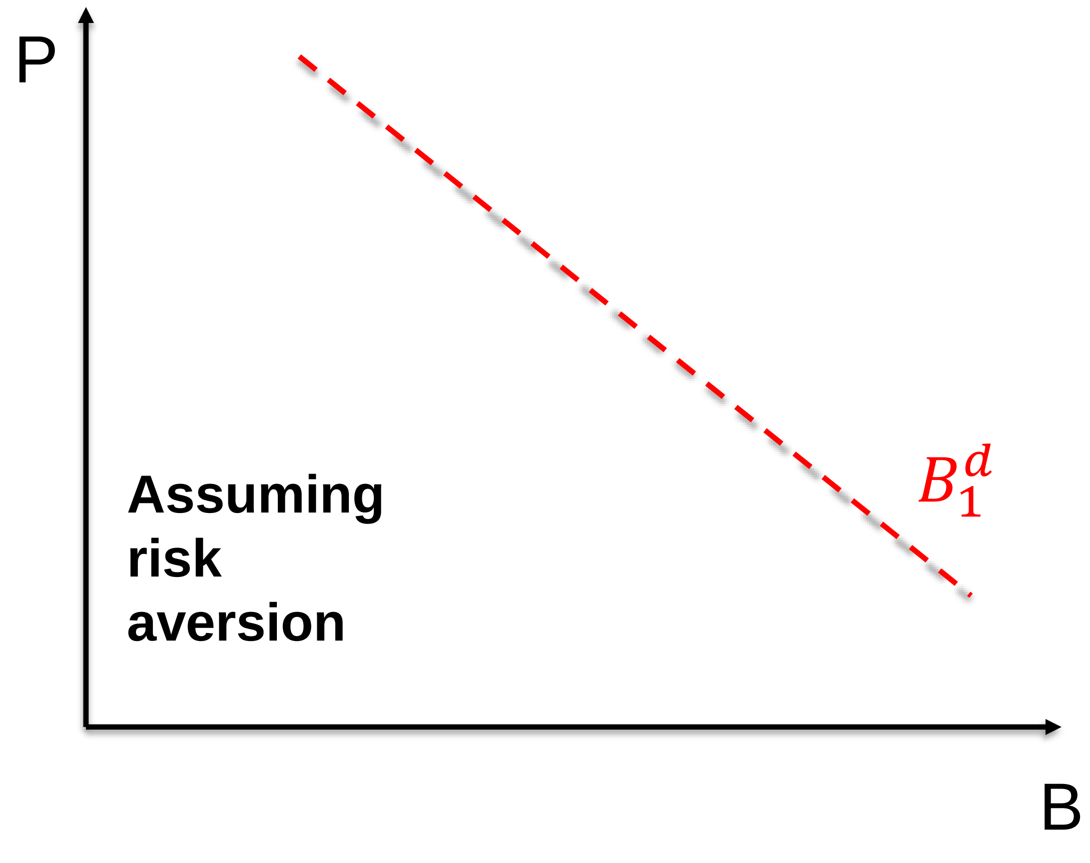
Q: What happens to 1-year discount bond with maturity of 1 year if $E(i)$ ↑?
A: Nothing! YTM=return, only today's *i* matters (in this case: $h = n$)

ECB announced increase of interest rates of 75 bps: $E(i_{t+1}) \uparrow$

Soon after announcement, expected interest rate of Italian 10ys Gov bonds ↑: $E(i_{t+1}) \uparrow$

Shift in Demand (3)

If riskiness of bonds (relative to other assets) ↑ ⇒ demand ? (relative)



Shift in Demand (3)

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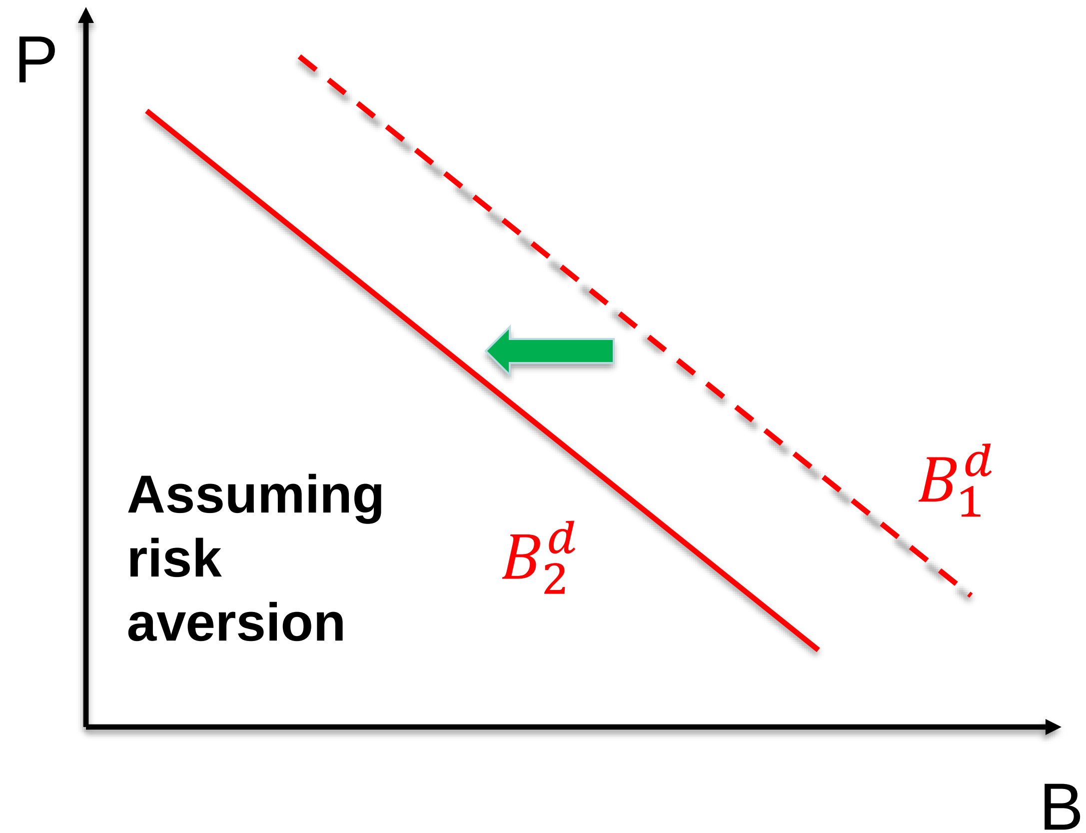
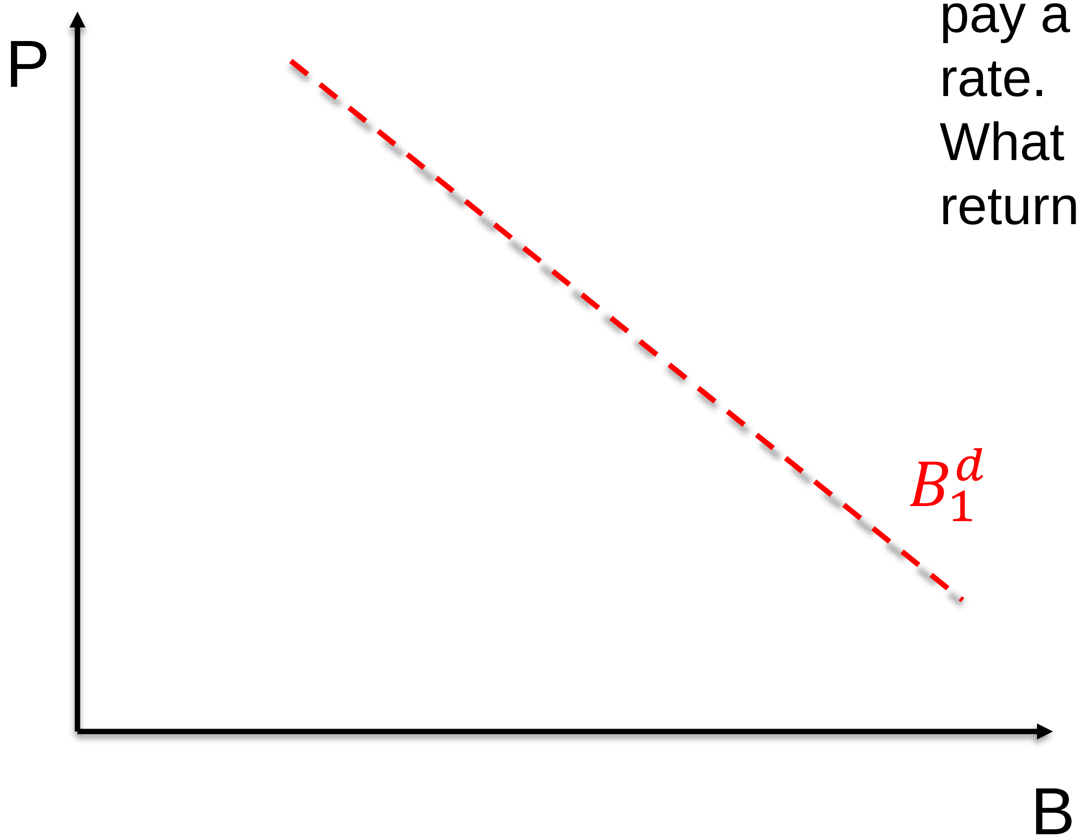


Shift in Demand (4)

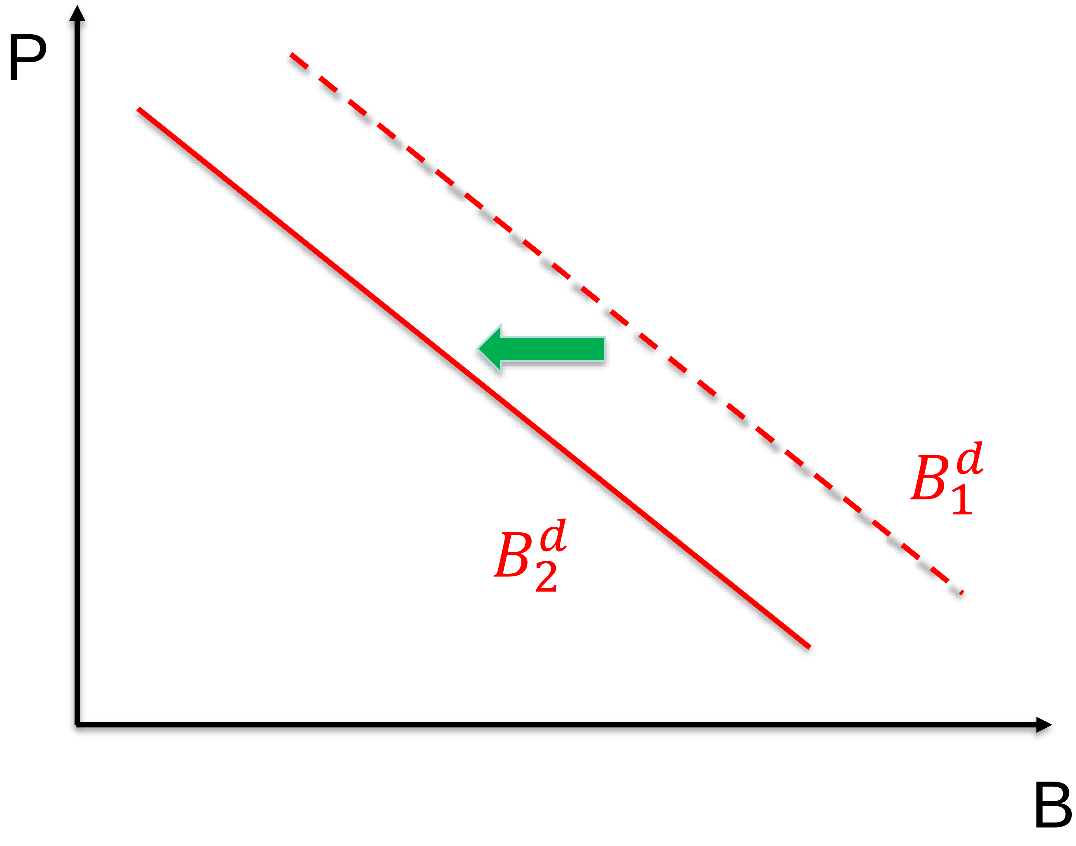
If $\pi^e \uparrow \Rightarrow$ demand ?
(absolute)



Hint: recall that bonds pay a nominal interest rate.
What happens to the real return?

Shift in Demand (4)

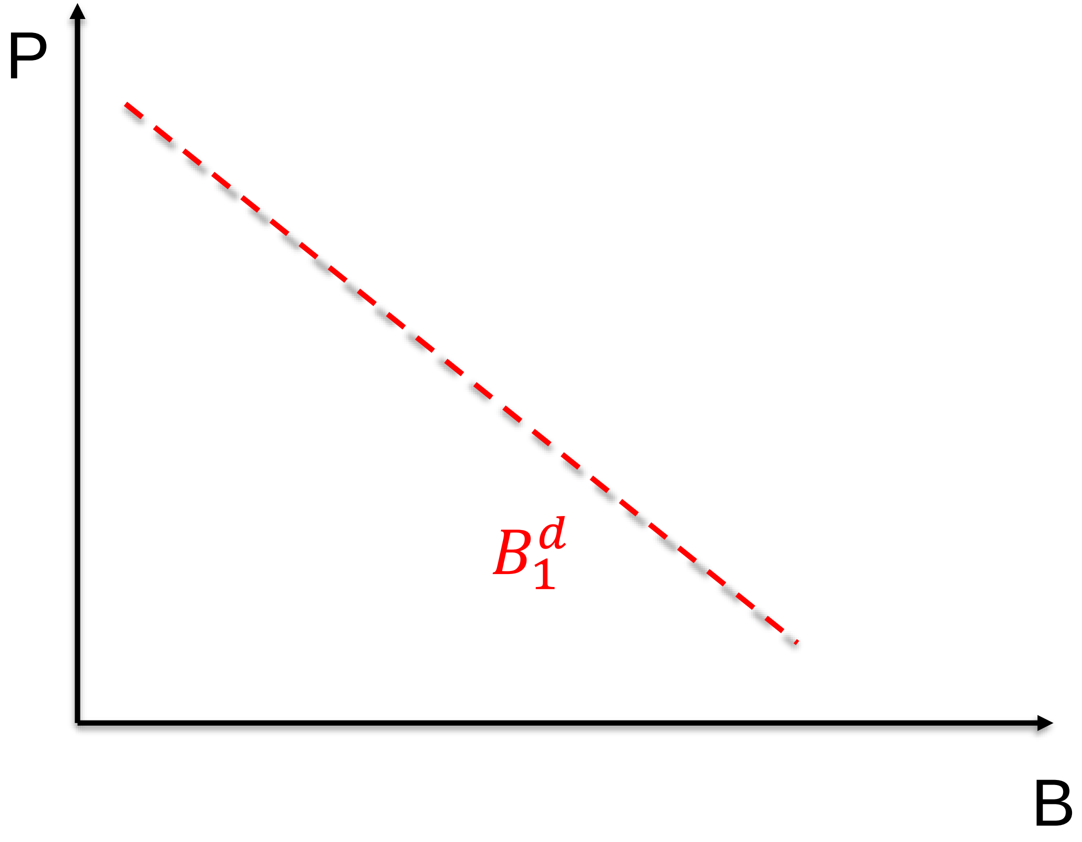
If $\pi^e \uparrow \Rightarrow$ demand $\downarrow$
(absolute)



Expected real return falls as inflation is expected to rise

Shift in Demand (5)

If liquidity of bonds ↑ ⇒ demand ?
(relative)



Shift in Demand (5)

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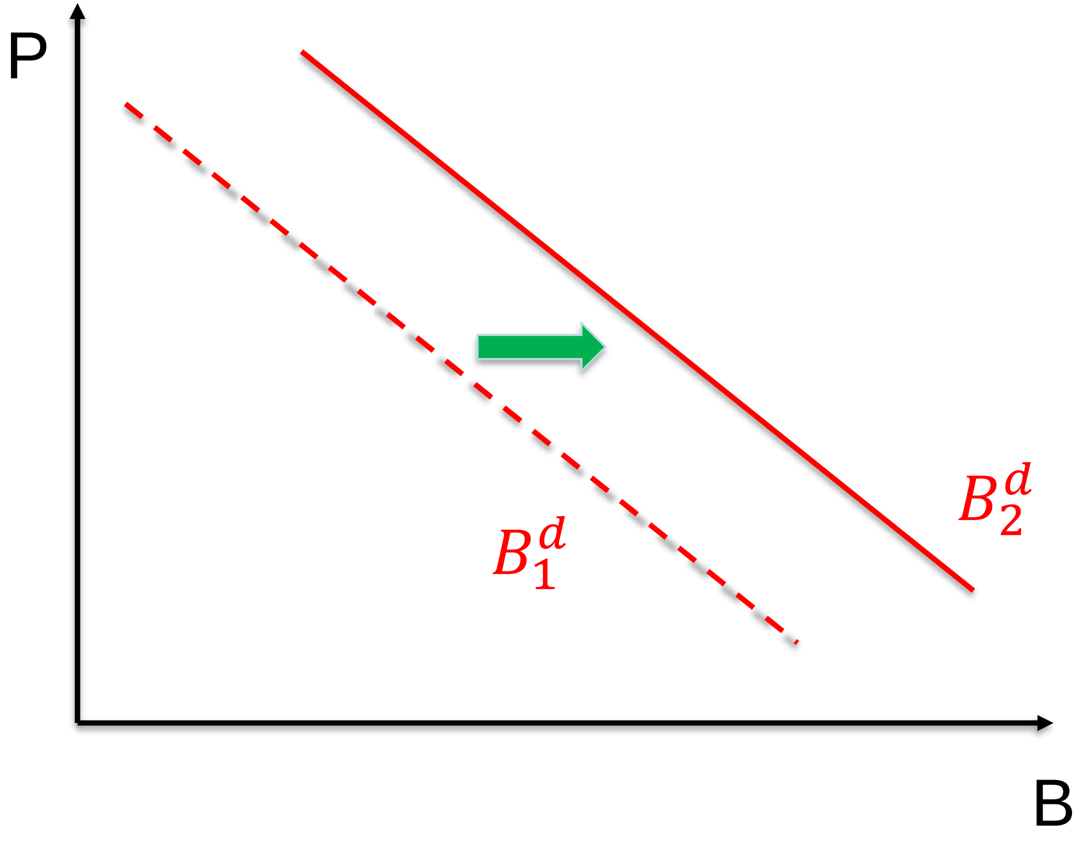


Shift in Supply

Now let's considers factors that shift the <u>supply</u> curve

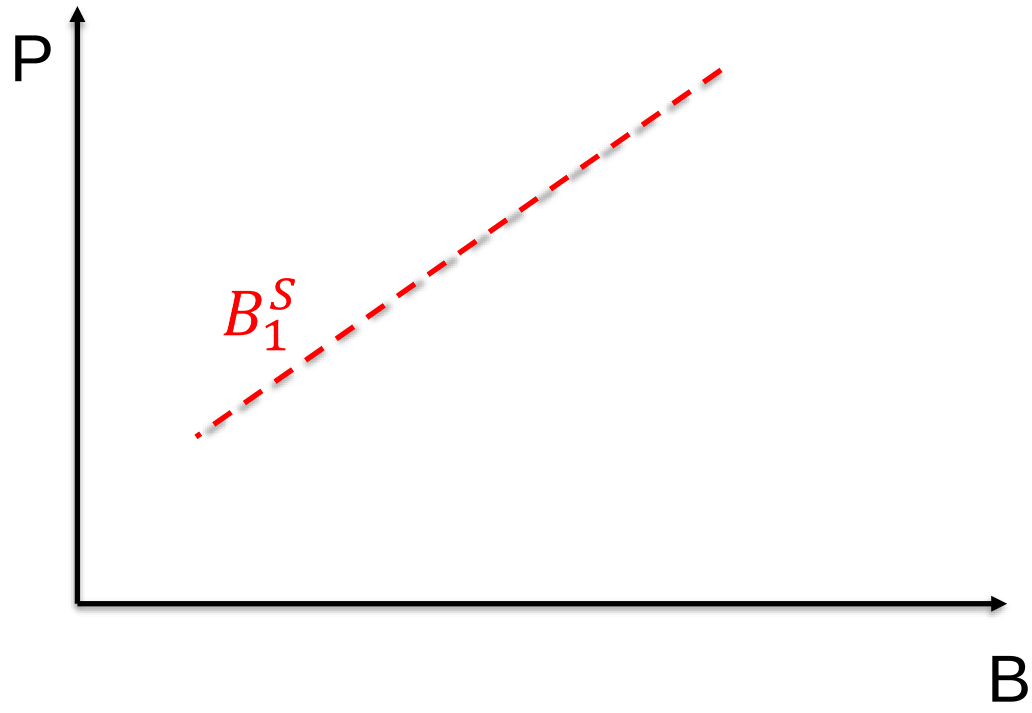
1. Expected **profitability** of investments (say productivity)

2. Expected future **inflation**: π^e

3. **Government deficit**

Shift in Supply (1)

If expected profitability of investment[(1)] ↑ ⇒ supply ? (absolute/relative)

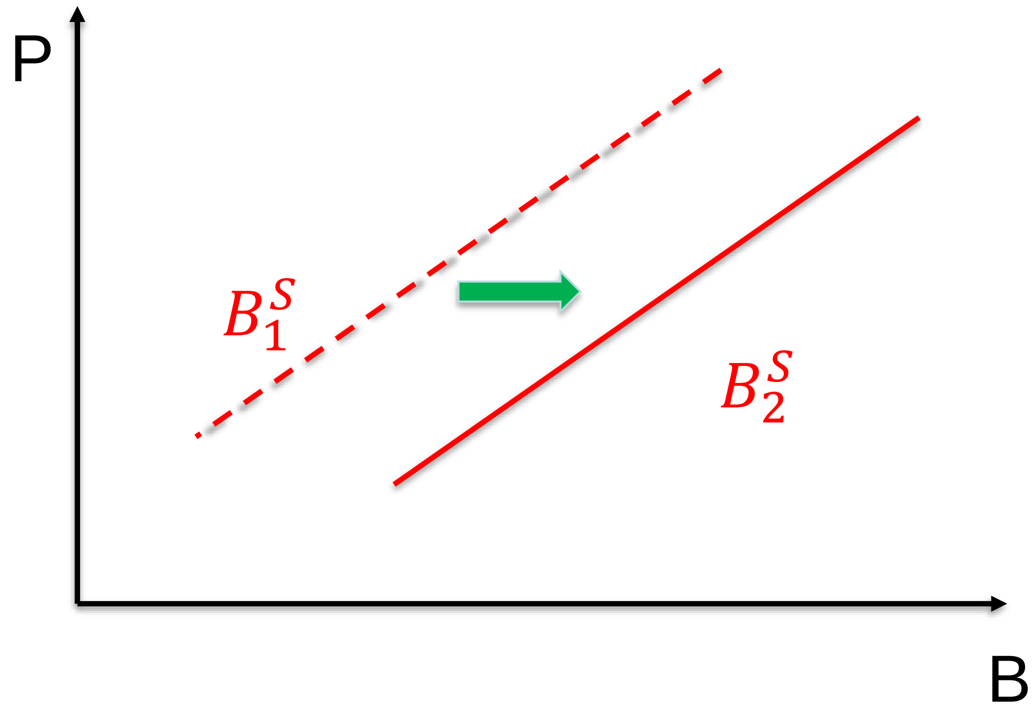


(1). profitability of investment may refer both to:

- a **micro** shock (firm investment opportunities increase due to, say, new patent) or
- a **macro** shock (boom/recession).

Shift in Supply (1)

If expected profitability of investment[1] ↑ ⇒ supply ↑ (absolute/relative)



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- a **micro** shock (firm investment opportunities increase due to, say, new patent) or
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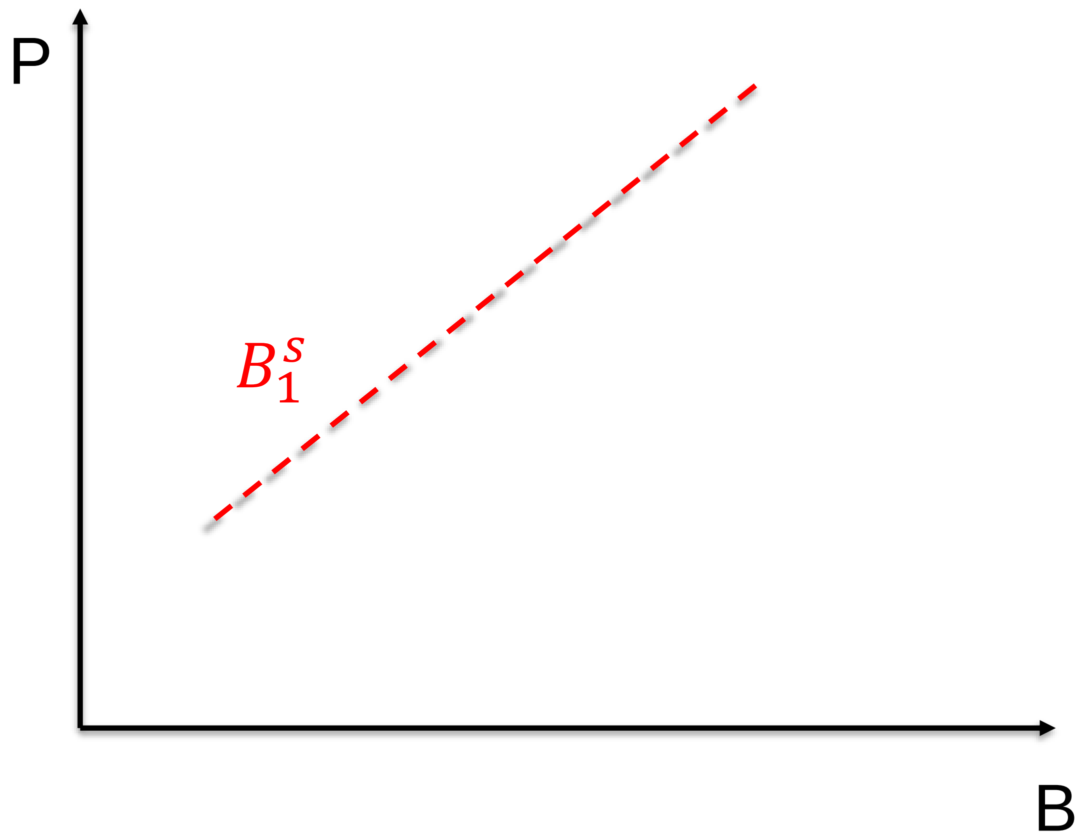
Shift in Supply (2)

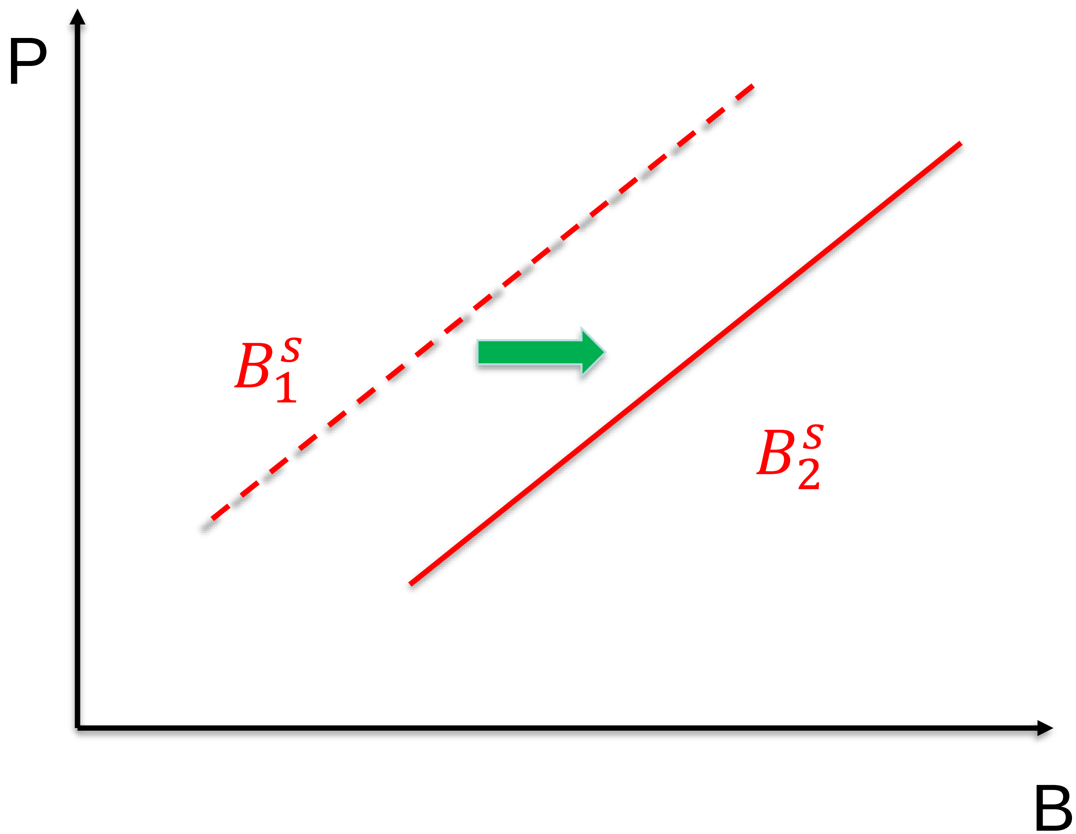
If $\pi^e \uparrow \Rightarrow$ supply ?
(absolute)



Shift in Supply (2)

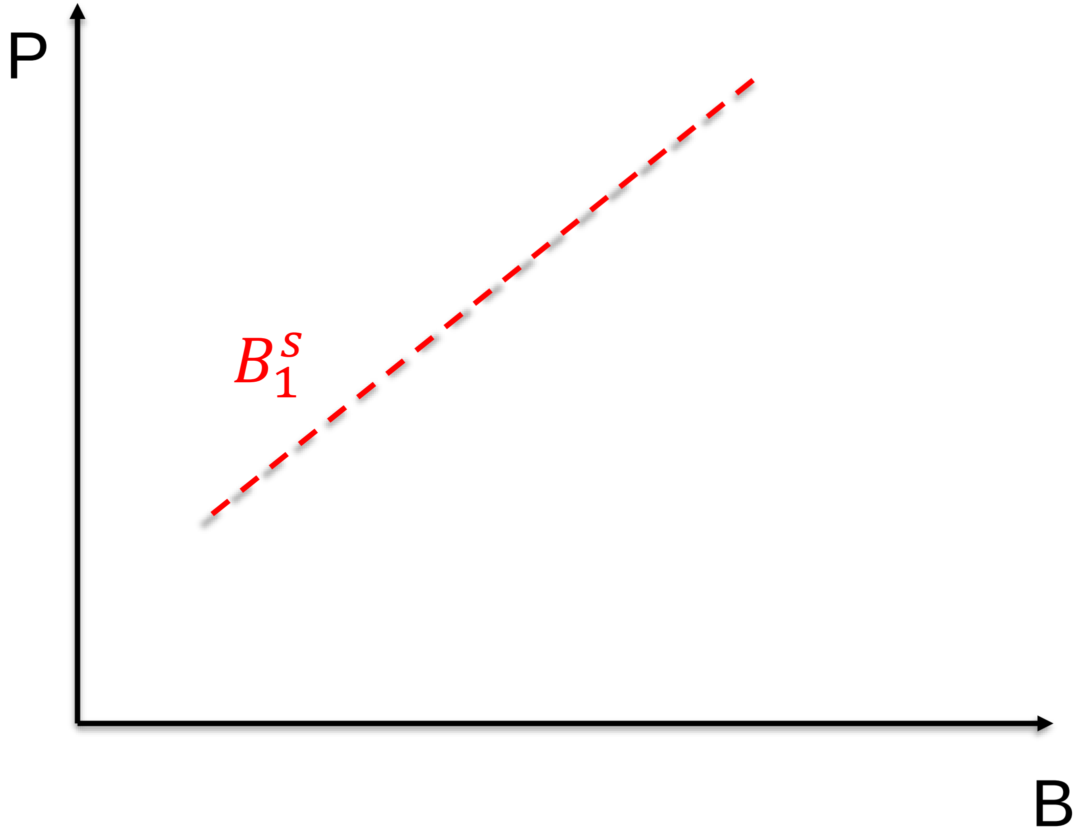
If $\pi^e \uparrow \Rightarrow$ supply $\uparrow$
(absolute)

Real cost of borrowing↓, it is less costly to borrow by issuing bonds ⇒ supply **increases**



Shift in Supply (3)

If government deficit ↑ ⇒ supply ?
(relative)



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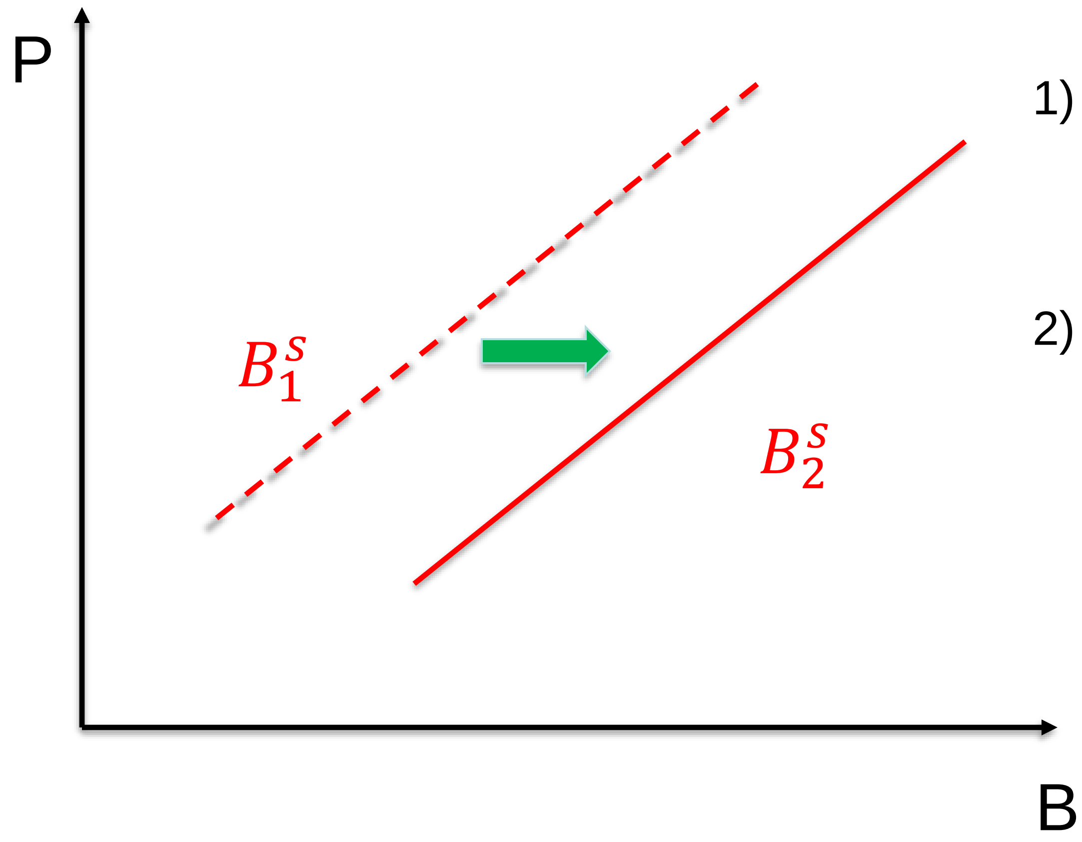
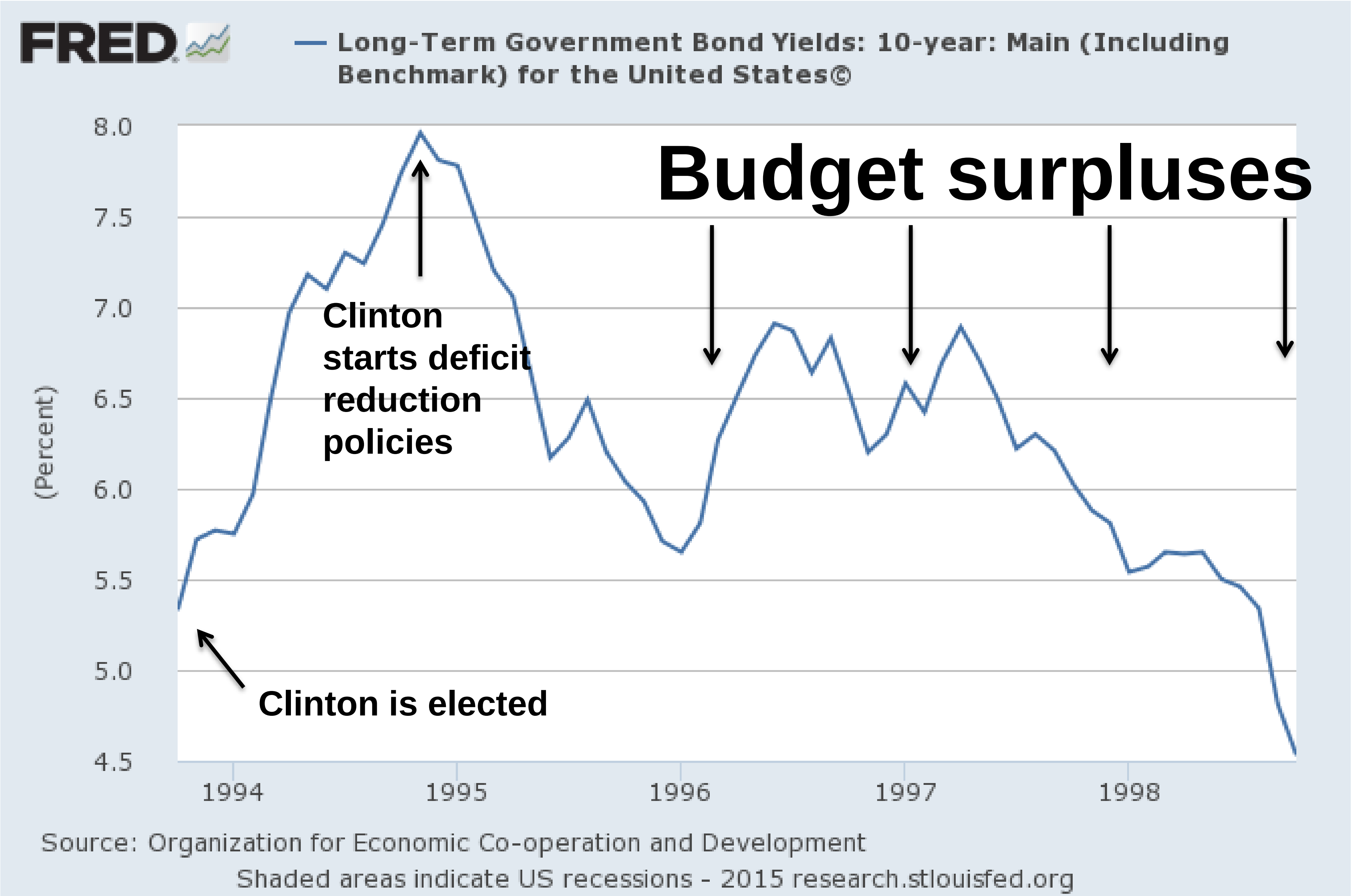


1) Larger deficit need more refinancing …
gov. issues more bonds

2) Interest rate required by investor is higher (so lower price) … recall crowding out in macro?

The Bond Market Vigilantes



Recap: shifts in Demand and in Supply

DEMAND

- ✓ if **wealth** (absolute*) ↑ ⇒ demand↑ (shift right)
- ✓ if **expected return** (relative*) ↑ ⇒ demand↑ (shift right)
- ✓ if risk (relative*) ↑ ⇒ demand↓ (shift left)
- ✓ if **expected inflation** ↑ (π^e↑; absolute*) ⇒ demand↓ (shift left)
- ✓ if liquidity of bonds (abs./rel.*) ↑ ⇒ demand↑ (shift right)

SUPPLY

- ✓ if **expected profit. of invest.** (abs./rel.*) ↑ ⇒ supply ↑ (shift right)
- ✓ if **expected inflation** ↑ (π^e ↑; absolute*) ⇒ supply ↑ (shift right)
- ✓ if government deficit ↑ (relative*) ↑ ⇒ supply ↑ (shift right)

(*): it is absolute if, 4 ex., a reform reduces trading commission costs to all securities; it is relative if it does so only to one category of bonds

Special case: the Fisher Effect

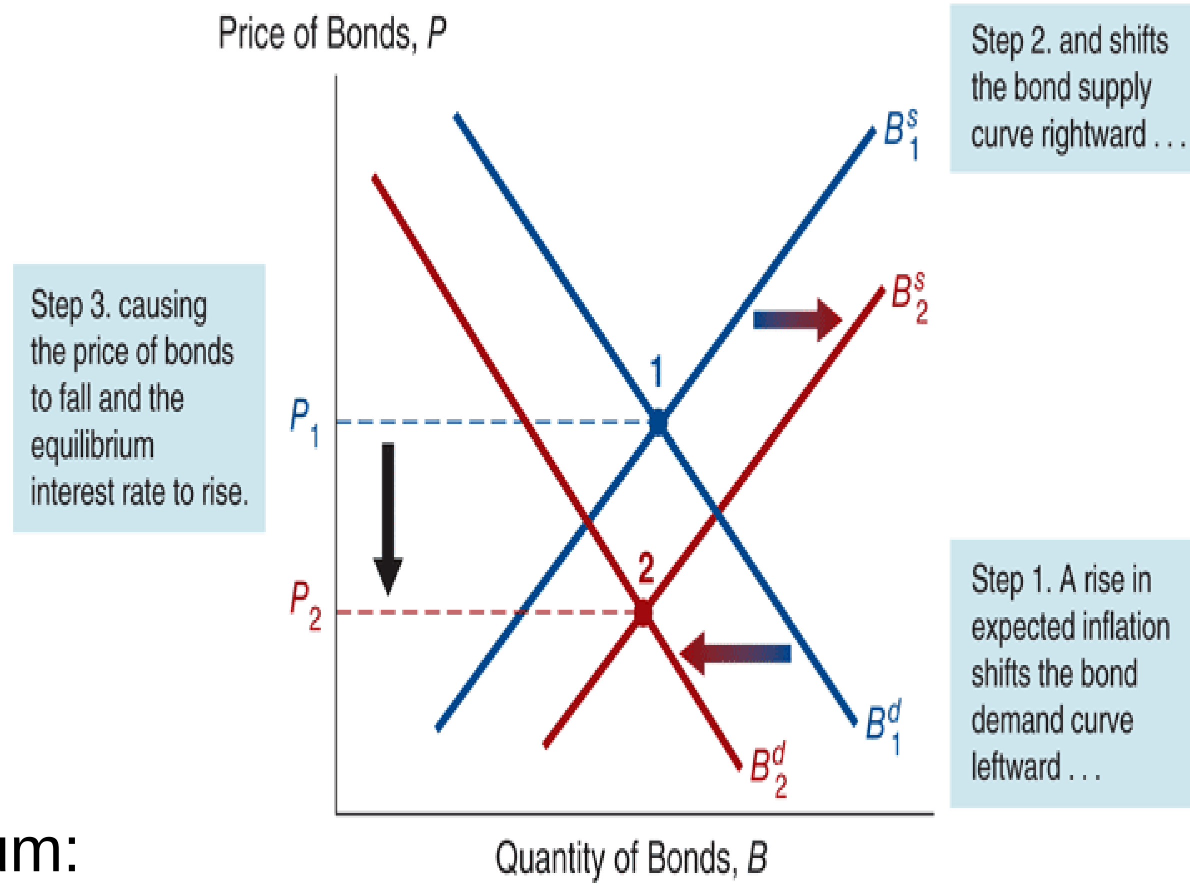
If $\pi^e\uparrow$, what happens to B^d and B^s ?

Fisher effect: *when expected inflation* ***rises****, interest rates will* ***rise***

Why?

Special case: the Fisher Effect

If $\pi^e\uparrow$, what happens to B^d and B^s ?



1) $R^e\downarrow$:
   B^d shifts in to left

2) real cost of issuance $\downarrow$:
   B^s shifts to right

3) In the new equilibrium:
   - $P\downarrow$, so $i\uparrow$
   - the quantity of bonds *may* change (both $\downarrow$ or $\uparrow$) or *may* <u>not</u> change

Figure 4.5 Expected Inflation and Interest Rates (Three-Month Treasury Bills), 1953–2013

Source: Expected inflation calculated using procedures outlined in Frederic S. Mishkin, "The Real Interest Rate: An Empirical Investigation," *Carnegie-Rochester Conference Series on Public Policy* 15 (1981): 151–200. These procedures involve estimating expected inflation as a function of past interest rates, inflation, and time trends. Nominal three-month Treasury bill rates from http://research.stlouisfed.org/fred2/.

Special case: Business Cycle Expansion

Suppose there is a **business cycle expansion**

Meaning national income increases

Q: What is the expected effect on equilibrium interest rates?

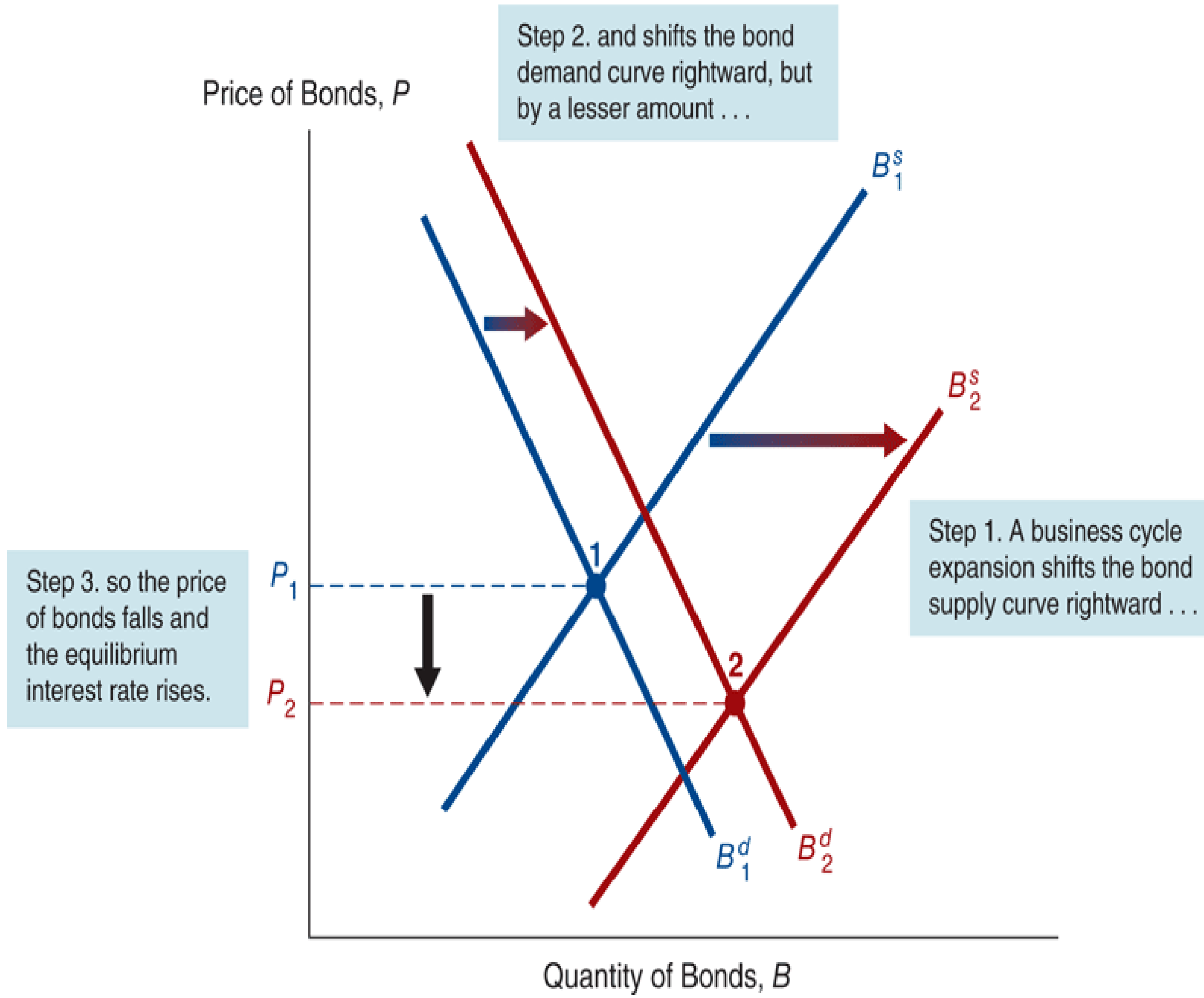
Hint: both demand and supply are affected. Indeed, if national income increases:

- wealth increases demand
- more business opportunities increase supply

Special case: Business Cycle Expansion

What happens to B^d and B^s ?

1. Expansion means wealth ↑, B^d↑ and it shifts out to right
2. Investment opportunities↑, B^s ↑ and it shifts out to right
3. **Ambigous total effect effect on *i*:**
   a) If B^s shifts more than B^d then P ↓, ***i*** ↑
   b) If B^s shifts less than B^d then P ↑, i ↓



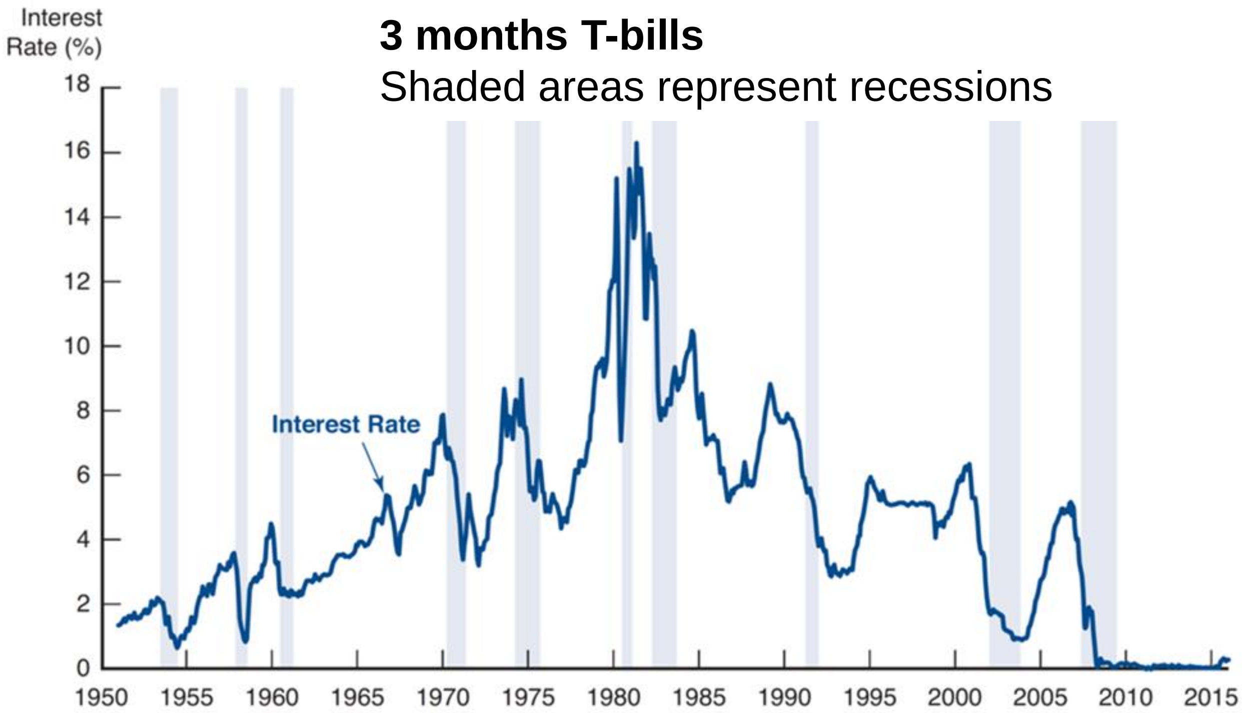
At the new equilibrium:
- quantity always increases
- intuitively, if B^s increases more than B^d, P↓ (i↑), and vice versa

Try at home to make graph of case *b)*.

Suggestive evidence on Business Cycles and Interest Rates

Interest rates are **positively** correlated with business cycle expansions (bond prices **negatively** correlated)

3 months T-bills
Shaded areas represent recessions



Special case: Negative Interest Rates

Why would anyone ever buy a government bond with **negative** yield?

- ✓ Safety (especially for large amounts, say >$100mil.)
- ✓ Makes sense if inflation is low and other investment opportunities are scarce
  - there are trillions (!) of $ of government debt yielding negative rates in the world
  - until recently, all German government bonds (including 30 year !!!) have had negative yield for relevant periods: see 10 year
  - in August 2021, 10-year common EU bonds had an interest of about -0.01% (like French debt)
  - situation dramatically changed after Covid crisis (global supply bottlenecks) and war in Ukraine … interest rates (and inflation) went on the rise, even German gov. bonds have, since then, a positive yield… no country nowadays

Special case: Negative Interest Rates

- ✓ In the wake of the global financial crisis, **interest rates** in Europe and the United States, as well as in Japan, have fallen to **extremely low levels**. How can we explain that within the framework discussed so far?
  - it's a little tricky, but we can do it!

- ✓ Inflation fell to negative rates, leading to lower real rates:
  - $B^d \uparrow$ (shifts out to right … less costly to borrow)
  - $B^s \downarrow$ (shifts out to left … less convenient to lend)
  - Net effect was an **increase** in *bond prices* (falling interest rates)

- ✓ Business cycle contraction led to:
  - $B^d \downarrow$ (shifts out to left … less need to borrow)
  - $B^s \downarrow$ (shifts out to left … less opportunities to lend)
  - Net effect was **not clear** on *bond prices*

- ✓ Overall:
  - the shift out to right in B^d is *less significant* than the shift out to left in B^d
  - so, the *net effect* was also an **increase in bond prices** ($P \uparrow$, $r \downarrow$)